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CMPG325 Phase 2: Client Implementation Review

Assigned Feature: Router-on-a-Stick Inter-VLAN Routing Implementation

1.Executive Summary

This report document is a formal submission for Phase2 of CMPG325. This report will provide information regarding how to configure,implement and verify the Router-on-a-Stick Inter-VLAN Routing process using Cisco Packet Tracer.

2. IP Addressing & VLAN Configuration Table

Device	Subnet/IP Address	Subnet Mask	Default Gateway	VLAN ID
CR1-PC	192.168.51.2	255.255.255.224	192.168.51.1	VLAN 10
CR2-PC	192.168.51.34	255.255.255.224	192.168.51.33	VLAN 20
CR3-PC	192.168.51.66	255.255.255.224	192.168.51.65	VLAN 30
CR4-PC	192.168.51.98	255.255.255.224	192.168.51.97	VLAN 40
CR5-PC	192.168.51.130	255.255.255.224	192.168.51.129	VLAN 50
Shared Printer	192.168.51.162	255.255.255.224	192.168.51.161	VLAN 60

3. Router-on-a-Stick Configuration Steps

Switch Trunking Setup:

Switch 1's configuration output establishes port GigabitEthernet0/1 as an operational trunk link. Verifies that the switch-to-router link is compatible with all VLANs (10–60).

Switch VLAN Assignment:

Switch1's show vlan short output, which displays access port mappings for every department: VLAN 10 (Fa0/1–2), VLAN 20 (Fa0/3), VLAN 30 (Fa0/4), VLAN 40 (Fa0/5), VLAN 50 (Fa0/6), and VLAN 60 (Fa0/7).

The screenshot shows the Cisco Packet Tracer interface with Router1 selected. The CLI window displays the following configuration:

```

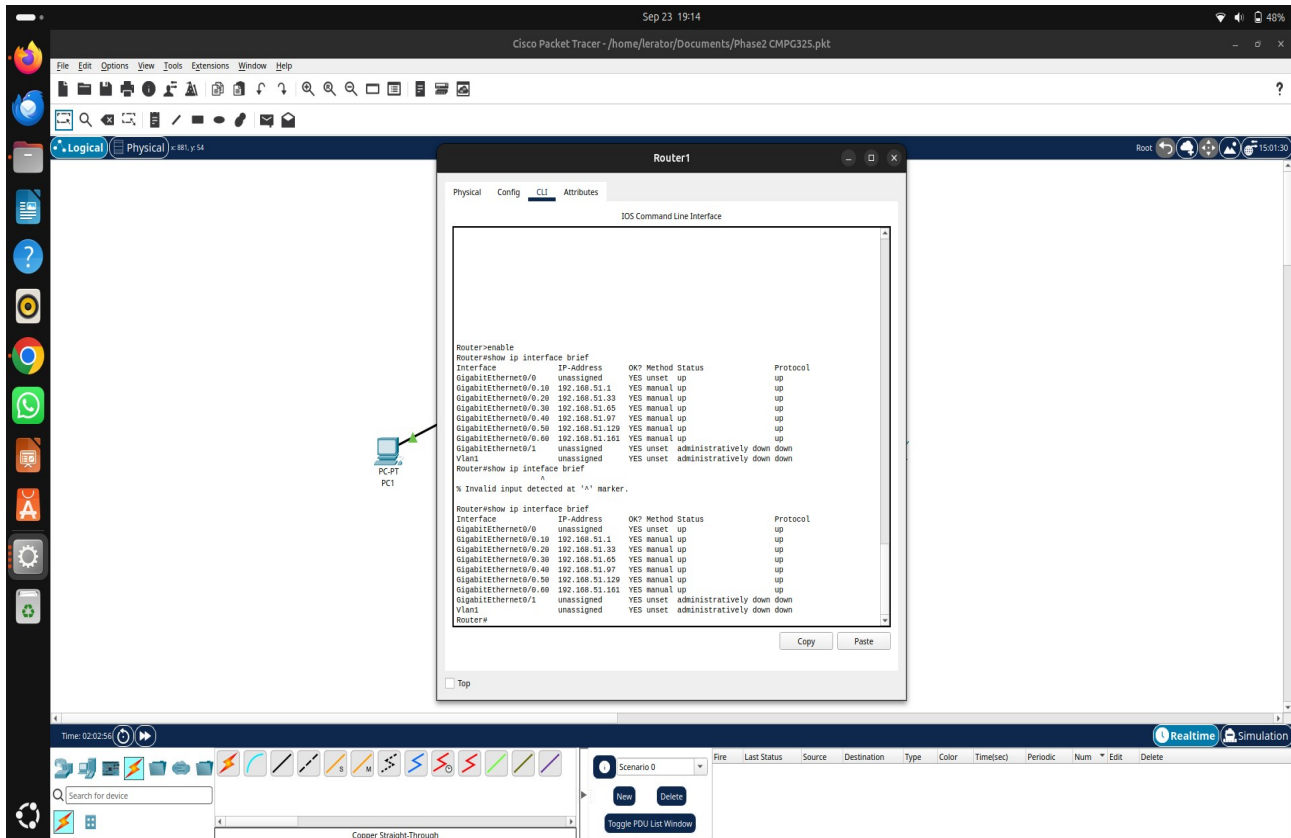
spanning-tree mode pvt
!
interface GigabitEthernet0/0
no ip address
duplex auto
speed auto
!
interface GigabitEthernet0/10
encapsulation dot1q 10
ip address 192.168.51.1 255.255.255.224
!
interface GigabitEthernet0/20
encapsulation dot1q 20
ip address 192.168.51.33 255.255.255.224
!
interface GigabitEthernet0/30
encapsulation dot1q 30
ip address 192.168.51.65 255.255.255.224
!
interface GigabitEthernet0/40
encapsulation dot1q 40
ip address 192.168.51.97 255.255.255.224
!
interface GigabitEthernet0/50
encapsulation dot1q 50
ip address 192.168.51.129 255.255.255.224
!
interface GigabitEthernet0/60
encapsulation dot1q 60
ip address 192.168.51.161 255.255.255.224
!
interface GigabitEthernet0/1
no ip address
duplex auto

```

The PC-PT PC1 is connected to the physical interface 0/0. The bottom of the window shows the Realtime Simulation mode and a toolbar for network configuration.

Router Running Configuration:

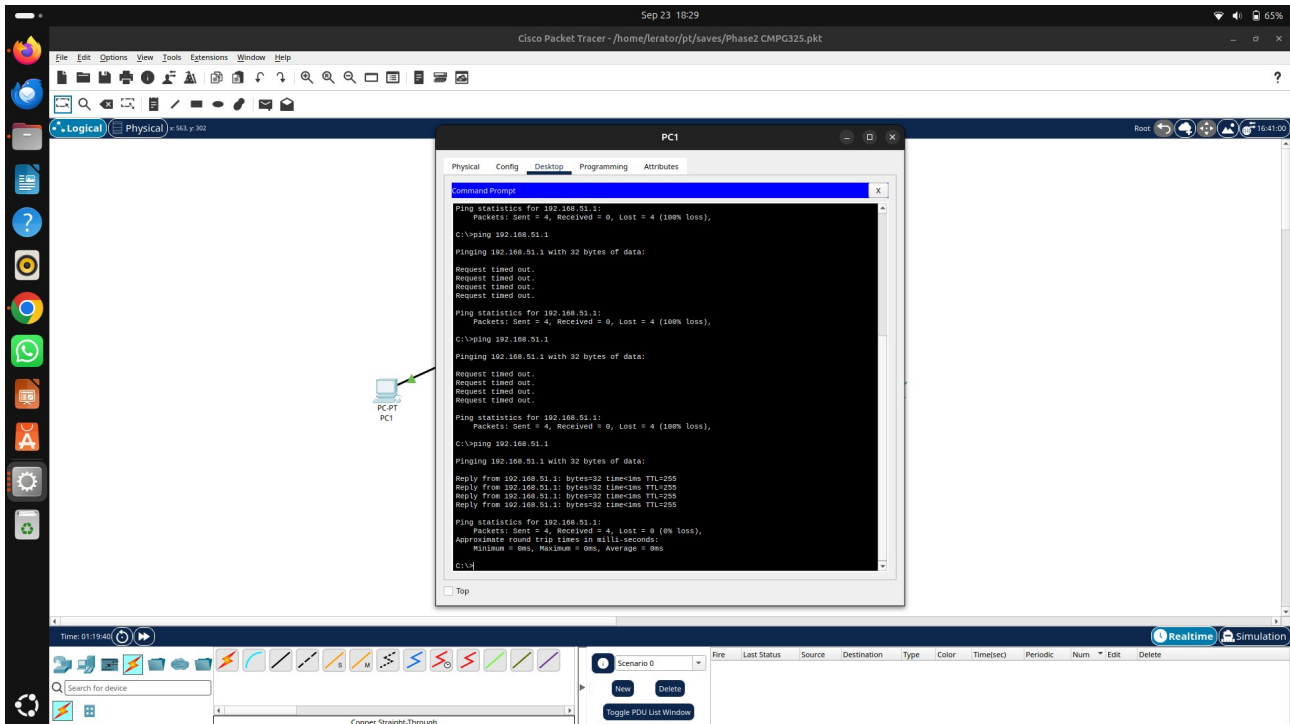
Router-on-a-Stick Inter-VLAN routing is achieved by implementing a configuration snippet on the router that shows active encapsulation tags (encapsulation dot1Q) and assigned IP subnets across all sub-interfaces.



4. Testing Router-On-A-Stick

4.1 Accessibility of the Gateway (192.168.51.1)

ICMP ping test from CR1-PC (192.168.51.2) to the local default gateway (192.168.51.1) on the GigabitEthernet0/0.10 sub-interface was successful.



4.2 Ping CR1 to CR2 Inter-VLAN (192.168.51.34)

From CR1-PC (VLAN 10) ping ICMP to CR2 (192.168.51.34, VLAN 20) is successful. Here is an example of active trunking and inter-vlan routing on the sub-interface GigabitEthernet0/0.20

```
PC1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.51.1
Pinging 192.168.51.1 with 32 bytes of data:
Reply from 192.168.51.1: bytes=32 time<1ms TTL=255
Reply from 192.168.51.1: bytes=32 time<1ms TTL=255
Reply from 192.168.51.1: bytes=32 time<1ms TTL=255
Reply from 192.168.51.1: bytes=32 time<1ms TTL=255
Ping statistics for 192.168.51.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>192.168.51.34
Invalid Command.
C:\>ping 192.168.51.34
Pinging 192.168.51.34 with 32 bytes of data:
Request timed out.
Reply from 192.168.51.34: bytes=32 time<1ms TTL=127
Reply from 192.168.51.34: bytes=32 time<1ms TTL=127
Reply from 192.168.51.34: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.51.34:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

4.3 Inter-VLAN Ping: 192.168.51.66 CR1 to CR3 and CR1 to CR4 (192.168.51.98)

ICMP ping test from CR1-PC (VLAN 10) to CR3 (192.168.51.66, VLAN 30) on switch port FastEthernet0/4 succeeded. confirms router-on-a-stick routing on the GigabitEthernet0/0.30 sub-interface.

CR1 to CR4: Inter-VLAN ping The ICMP ping test from CR1-PC (VLAN 10) to CR4 (192.168.51.98, VLAN 40) on the switch's FastEthernet0/5 port was successful. Confirms packet forwarding on the router subinterface.

Sep 29 14:05 Cisco Packet Tracer - /home/lerator/Documents/Phase2 CMPG325.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical 561, 583

PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 192.168.51.66

Pinging 192.168.51.66 with 32 bytes of data:

Request timed out.
Reply from 192.168.51.66: bytes=32 time=1ms TTL=127
Reply from 192.168.51.66: bytes=32 time<1ms TTL=127
Reply from 192.168.51.66: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.51.66:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.51.98

Pinging 192.168.51.98 with 32 bytes of data:

Request timed out.
Reply from 192.168.51.98: bytes=32 time<1ms TTL=127
Reply from 192.168.51.98: bytes=32 time<1ms TTL=127
Reply from 192.168.51.98: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.51.98:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.51.130

Pinging 192.168.51.130 with 32 bytes of data:

Request timed out.
Reply from 192.168.51.130: bytes=32 time<1ms TTL=127
Reply from 192.168.51.130: bytes=32 time<1ms TTL=127
Reply from 192.168.51.130: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.51.130:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PC-PT PC1

Time: 00:47:59

Scenario 0

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
------	-------------	--------	-------------	------	-------	-----------	----------	-----	------	--------

Toggle PDU List Window

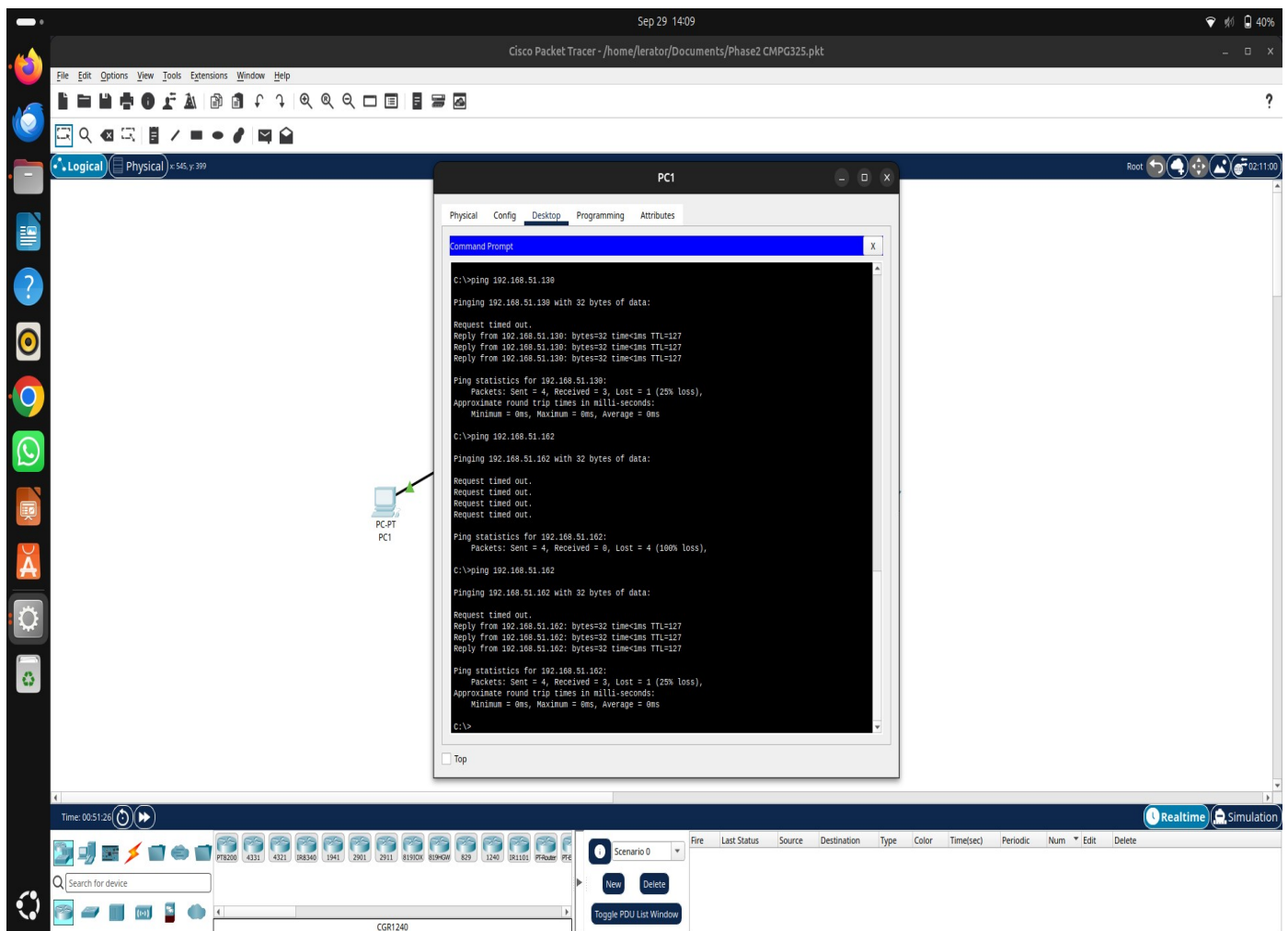
CGR1240

4.4 Inter-VLAN Ping: 192.168.51.130 CR1 to CR5

ICMP ping test from CR1-PC (VLAN 10) to CR5 (192.168.51.130, VLAN 50) switch port FastEthernet0/6 succeeded. Checks reachability through the router subinterface GigabitEthernet0/0.50.

CR1 to Shared Printer Inter-VLAN Ping (192.168.51.162)

ICMP ping test from CR1-PC (VLAN 10) to Shared Printer (192.168.51.162, VLAN 60) on switch port FastEthernet0/7 was successful. Confirms full inter-VLAN connectivity to the shared resource on the GigabitEthernet0/0.60 subinterface



5. Troubleshooting and Verification

When the system was being deployed and tested, certain early problems with connectivity were found during testing of the communication between devices in different VLANs. These issues were solved through carrying out configuration checks and additional testing.

We pinged the shared printer and the other departmental PCs from CR1-PC to check the network, and we also pinged the default gateway. The fact that the ping replies were successful proved that the Router-on-a-Stick configuration was working and that the VLANs we had tested were able to communicate with one another.

6. Conclusion

The project was a success in implementing Router-on-a-Stick Inter-VLAN routing on the Northwest Cricket Union network using Cisco Packet Tracer. It involved the following steps: configuring the VLANs on the switch, assigning the access ports to the relevant VLANs, configuring trunking between the switch and the router, and setting up router sub-interfaces with the required IP addresses.

The test of connectivity was carried out in order to check the communication between the CR1-PC, the departmental PCs in the other VLANs, the default gateway and the shared printer, and successful ping tests proved that the network as configured was permitting communication between the VLANs that had been tested.

The project provided hands-on experience in setting up VLANs, routing between VLANs, troubleshooting networks, and checking connectivity.

7. References

- Cisco Networking Academy. *Cisco Packet Tracer*. <https://www.netacad.com/cisco-packet-tracer>
- North-West University. (2026). *CMPG325 Computer Networks: Semester Project Brief*.